

MATH9102: Fundamentals of Data Analysis

Unit 5: Comparing More Than Two Groups
ANOVA, Kruskal–Wallis and Post-Hoc Tests

Today

1. **Before You Test** — why many t -tests fail, and the map of tests
2. **How ANOVA Works** — variance, F , and what an omnibus test can say
3. **The Worked Example** — optimism across three age groups, end to end
4. **Many Comparisons** — when to follow up, and how to correct
5. **When Normality Fails** — Kruskal–Wallis and Dunn's test
6. **Reporting**

Same six steps as Unit 4. Step 4 now has two halves.

Important Note

The omnibus test tells you **that** the groups differ. Only the follow-up tells you **where** — and only if you control the error rate while you look.

Before You Test

Where This Week Sits

Unit 4: two groups. This week: three or more.

What stays the same

- The question — do the groups differ?
- The six steps, and the work in steps 2, 3, 5 and 6
- Welch by default; normality checked within groups
- An effect size beside every test

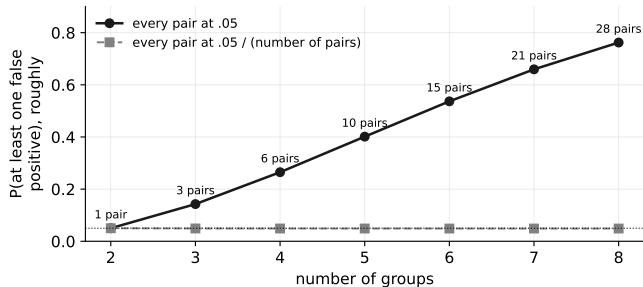
What is new

- One test of all the groups at once — the *omnibus*
- A second step to find *which* groups differ
- A correction for looking many times
- Effect sizes as shares of variance

Continuous outcome, categorical grouping

The grouping variable now has three or more levels: age bands, education, social class, treatment arms. The outcome is still a measurement — or, in the second half, a rank.

Why Not Just Run Lots of t -Tests?



Three groups make three pairs; six make fifteen. Each t -test at .05 carries its own one-in-twenty chance of a false positive, and the chances accumulate.

The familywise error rate

The chance of *at least one* false positive in the family of tests. By six groups it is worse than a coin toss.

The Map of Multi-Group Tests

The same two questions as Unit 4: how are the groups formed, and what kind of variable is the outcome?

	Independent samples different people in each group	Related samples the same people, three or more times
Interval / parametric	One-way ANOVA (Welch's)	Repeated-measures ANOVA
Ordinal / non-parametric	Kruskal–Wallis	Friedman's ANOVA

This week

The left-hand column. The right-hand column is named so you recognise a repeated-measures design when you have one.

Vocabulary. The grouping variable is the **factor**; its categories are its **levels**. One factor makes a *one-way* ANOVA, however many levels it has.

The Six Steps, With One Addition

Unit 4's procedure, unchanged except at step 4.

1. **State** the hypothesis pair
2. **Inspect** the variables
3. **Check** the assumptions, and decide
4. **Run** the omnibus test
then, if it finds something, the **post-hoc** comparisons
5. **Report** the result, with an effect size
6. **Conclude** about the hypothesis

The order is the point

The post-hoc step answers a question the omnibus test has already said is worth asking. Run it first and you are back to many t -tests.

How ANOVA Works

Analysis of Variance, to Compare Means

The name describes the method, not the question. The question is about means.

The idea

- Measure how much the scores vary *within* each group
- Work out how much the group means would vary from one another by chance alone, given that
- Compare it with how much they actually vary

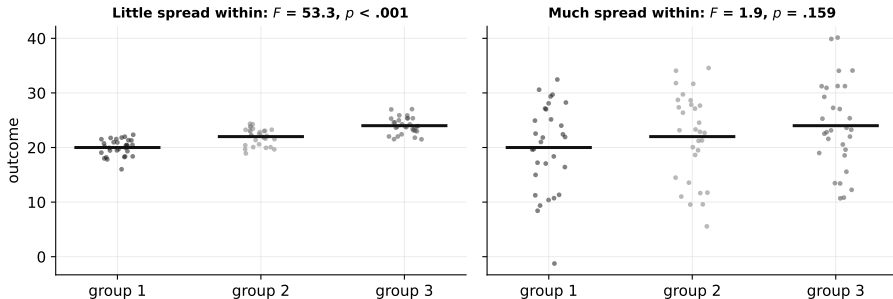
The verdict

- Means further apart than chance explains: evidence that they differ
- Means about as far apart as chance explains: no evidence
- It is the t -test's logic, for any number of groups

With two groups, they are the same test

A one-way ANOVA on two groups gives $F = t^2$ and the same p . ANOVA is the general case.

Two Sources of Variation



The group means are **identical** in the two panels.
Only the spread inside each group differs.

What the test sees

Left: the means stand out from the noise.
Right: the same gaps are lost in it.
Whether means “differ” depends on the spread around them.

Sums of Squares, and Where F Comes From

$$\underbrace{\sum (x - \bar{x})^2}_{SS_{\text{total}}} = \underbrace{\sum n_j (\bar{x}_j - \bar{x})^2}_{SS_{\text{between}}} + \underbrace{\sum (x - \bar{x}_j)^2}_{SS_{\text{within}}}$$

$$F = \frac{MS_{\text{between}}}{MS_{\text{within}}} = \frac{SS_{\text{between}}/(k - 1)}{SS_{\text{within}}/(N - k)}$$

- **Between:** how far each group mean sits from the grand mean
- **Within:** how far each score sits from its own group's mean
- Dividing by degrees of freedom turns each into a *mean square*

Two sets of degrees of freedom

$k - 1$ for the groups and $N - k$ for the people.
Both are reported: $F(k - 1, N - k)$.

Reading the F -Ratio

F is the signal divided by the noise — the variation the groups explain, over the variation they leave.

- If H_0 is true, both mean squares estimate the same thing, and F sits near 1
- An F well above 1 means the means vary more than chance explains
- An F below 1 can never be significant: the groups vary *less* than chance would make them

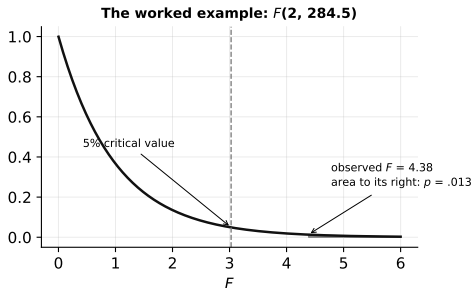
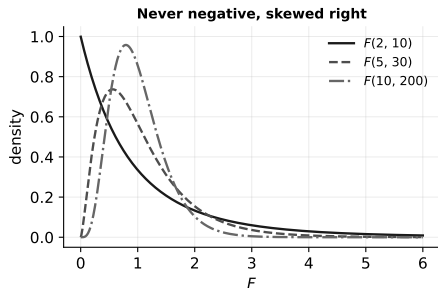
Reporting convention

F , both degrees of freedom, and p :

$$F(df_1, df_2) = \dots, \quad p = \dots$$

Two degrees of freedom, always: $k - 1$, then the second. The second need not be a whole number — see Welch.

The F Distribution



- Never negative: it is a ratio of two variances
- Skewed to the right, with a different curve for every pair of degrees of freedom
- As both grow it narrows and becomes more symmetric around 1

Always one-tailed

Only a large F is evidence against H_0 , so p is the area to the right. The direction of a difference comes from the means, not from F .

An Omnibus Test

ANOVA asks one question of all the groups at once.

Hypotheses

H₀: $\mu_1 = \mu_2 = \dots = \mu_k$

H₁: at least one mean differs from the others

What H₁ does not say

Not that *every* group differs from *every* other. One group standing apart is enough to make H₀ false.

So a significant F tells you that, not where

It establishes that the means are not all equal. Which ones differ is a second question, answered by post-hoc comparisons that keep the familywise error rate in check.

Assumptions of One-Way ANOVA

What it assumes

1. The outcome is measured at least at interval level
2. The outcome is roughly normal **within each group**
3. The groups have similar variances
4. Observations are independent of one another

How much each one matters

- (1) is your judgement, not a test
- (2) matters least when the groups are large and similar in size
- (3) we stop assuming — Welch's ANOVA, as with the t -test
- **(4) has no repair.** It is fixed by the design

Watch out

Unequal group sizes *and* unequal variances together are where the classic F goes wrong — in either direction. Welch's version is built for exactly that case.

The Worked Example

Step 1: A Worked Question

Hypotheses

H₀: Mean optimism is the same in all three age groups.

H₁: Mean optimism differs in at least one age group.

Outcome

- toptim — Life Orientation Test total
- Six items, so it runs from 6 to 30

Factor

- agegp3 — three age bands
- 18–29, 30–44, 45 and over

Decision rule

- $\alpha = .05$, fixed now
- Follow up only if the omnibus test is significant

Stand Up — 5 minutes

Five minutes. The question is on the table; the data are not.

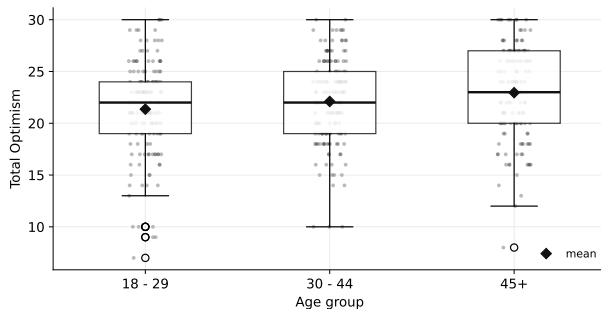
Food for thought

- Will age matter for optimism? Which way?
- If it does, will all three groups differ, or just the two ends?
- How large a difference, on a scale from 6 to 30, would you call worth reporting?

Say it out loud

Tell the person next to you, or jot it on paper — not on a screen. A prediction made before the output is a prediction. One made afterwards is a story.

Step 2: Look Before You Test



What to read off

- The means edge upward with age
- The boxes overlap almost entirely
- Every group bunches at the top of the scale

Before any test

Boxes this similar promise a small effect, whatever p turns out to be.

Step 2: Describe Each Group

```
# Step 2. The grouping variable, then the outcome within each group.  
# Sorting is not an order: say what the order is once, and reuse it.  
ages = ["18 - 29", "30 - 44", "45+"]  
m9.frequency(survey, "agegp3")  
m9.describe_by(survey, "toptim", "agegp3")  
  
# What a complete-case analysis costs, on the model variables only.  
missing = m9.missingness(survey, ["toptim", "agegp3"])  
print(missing.attrs["summary"])
```

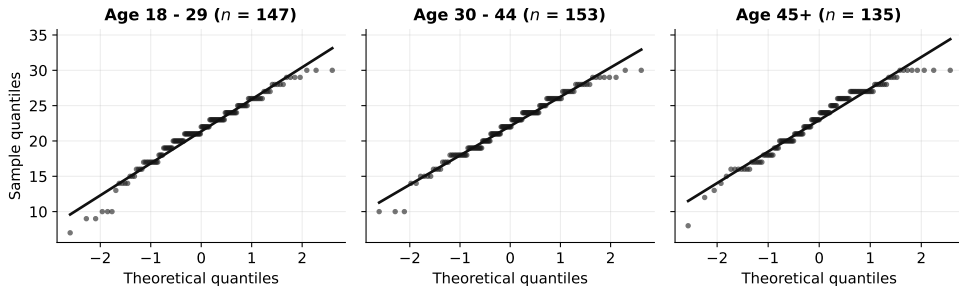
You need, per group

- n , and how many are missing
- Mean and standard deviation
- Enough shape to justify step 3

Watch out

Sorting is not an order. Some of this week's groupings sort with the youngest group *last*. Write the order out once, and reuse it.

Step 3: Normality, Within Each Group



On the line through the middle, flattening at the top: a **ceiling**, where respondents sit on the maximum score. Over a hundred people per group, and nothing worse than that.

The decision

A parametric test is defensible. Say that it is a judgement, and say which plots it was made from.

Step 3: Equal Variances, Described, Not Tested

```
# Step 3. Describe the spread in each group. Do not branch on it.
opt = survey[["toptim", "agegp3"]].dropna()
groups = [opt.loc[opt.agegp3 == a, "toptim"] for a in ages]

sds = [g.std(ddof=1) for g in groups]
print(f"largest SD / smallest SD = {max(sds) / min(sds):.2f}")

levene = stats.levene(*groups, center="median")
print(f"Levene (median-centred): p = {levene.pvalue:.3f}")
```

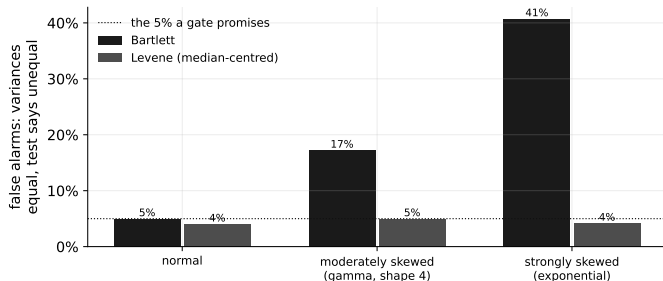
Our convention

- Welch's ANOVA, always
- Report the group SDs and their ratio
- Look at Levene's test, and branch on nothing

Why not gate?

Choosing the test from another test's result distorts the error rate of the one you report — and $p = .049$ against $p = .051$ would give two analyses of one dataset.

Bartlett Sees Skew as Spread



Three groups drawn from *one* population, so the variances are equal and every rejection is a false alarm. On skewed data, Bartlett's test raises far more alarms than the five per cent it promises.

Watch out

You will meet Bartlett's test used to choose an ANOVA. It mistakes non-normality for unequal variances. Median-centred Levene does not.

Step 4: Welch's F , and the Classic F

```
# Step 4. Welch is the default. The classic F is shown only for comparison -  
# never choose between them on the outcome of a variance test.  
welch = stats.f_oneway(*groups, equal_var=False)  
classic = stats.f_oneway(*groups)  
  
print(f"Welch: F = {welch.statistic:.3f}, p = {welch.pvalue:.4f}")  
print(f"Classic: F = {classic.statistic:.3f}, p = {classic.pvalue:.4f}")  
  
# scipy gives F and p. pingouin also prints Welch's degrees of freedom.  
pg.welch_anova(data=opt, dv="toptim", between="agegp3")
```

On this example

The two agree in verdict. That is the usual case, and it is why defaulting to Welch costs almost nothing — while protecting you in the case where it would have mattered.

What Welch Changes

How it works

- Each group is weighted by how precisely its mean is known: n_j/s_j^2
- A small or very spread-out group counts for less
- The second degrees of freedom are adjusted down to match — and stop being a whole number

What it costs

- A little power, when variances really are equal and groups really are balanced
- Nothing in the verdict here

Watch out

scipy's `f_oneway(..., equal_var=False)` returns F and p but no degrees of freedom. Take them from `pg.welch_anova`, or from a generated report — never as $N - k$, which belongs to the classic test.

Step 4b: Where Is the Difference?

Post-hoc comparisons test every pair, while holding the familywise error rate at five per cent.

Games–Howell

- Does not assume equal variances
- The follow-up that matches Welch's ANOVA
- **Our default**

Tukey's HSD

- Assumes equal variances
- The one most textbooks show
- The partner of the classic F

Match the follow-up to the omnibus

You meet procedures that run Bartlett's test to choose between Tukey and Games–Howell. That is the variance gate again, one step later. Choose Games–Howell for the same reason you chose Welch.

Games–Howell, and Tukey

```
# Step 4b. Where is the difference? Games-Howell does not assume equal
# variances, so it is the follow-up that matches Welch's F.
gh = pg.pairwise_gameshowell(data=opt, dv="toptim", between="agegp3")
gh[["A", "B", "diff", "pval", "hedges"]]

# Tukey's HSD assumes equal variances. Shown for comparison only.
print(stats.tukey_hsd(*groups))
```

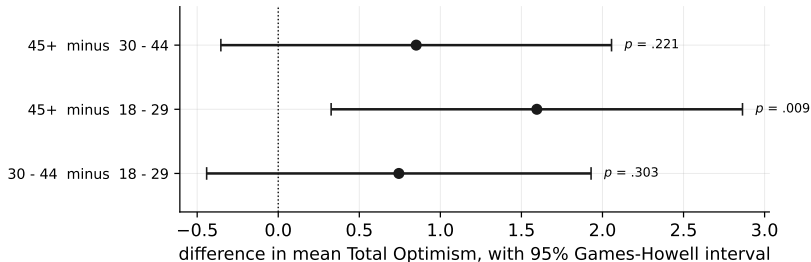
Read, per row

- Which pair, and which way: diff
- The adjusted p : pval
- The size of that one difference: hedges

Adjusted already

Both procedures build the correction into their p -values. Compare pval with .05 directly — do not divide again.

Reading the Post-Hoc Comparisons



One interval clears zero: the youngest group against the oldest. The middle group cannot be told apart from either neighbour.

Not a contradiction

The omnibus test asks one question of all the data. Each pair asks a narrower question of two-thirds of it. Significant overall, one pair significant, is common.

Step 5: Effect Size — η^2 and ω^2

$$\eta^2 = \frac{SS_{\text{between}}}{SS_{\text{total}}}$$

$$\omega^2 = \frac{SS_{\text{between}} - (k - 1) MS_{\text{within}}}{SS_{\text{total}} + MS_{\text{within}}}$$

Eta squared — the share of the variation in the outcome that the groups account for *in this sample*. Always a little too large: the group means were fitted to the same data.

Omega squared — corrects that, estimating the share in the *population*. Describe this one in words.

Benchmarks

.01 small, .06 medium, .14 large — for both.

Benchmarks, not verdicts. With groups in the hundreds, η^2 and ω^2 are close; with groups of ten they are not.

Two Effect Sizes, One to Describe

```
# Step 5. Eta squared is this sample's share of the variance. Omega squared
# estimates the population's, and is the one to describe in words.
eta2 = m9.eta_squared(groups)
omega2 = m9.omega_squared(groups)

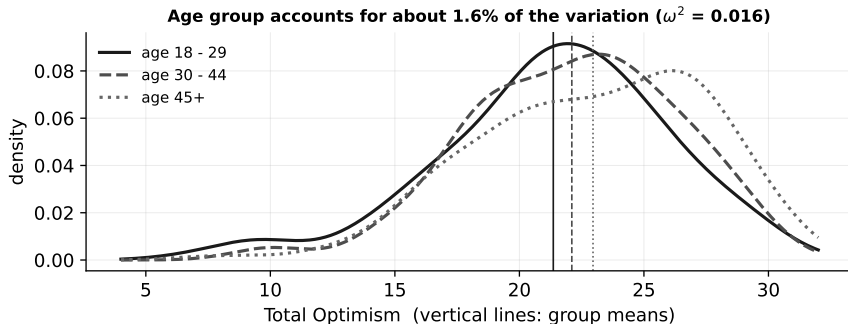
print(f"eta squared = {eta2:.3f}")
print(f"omega squared = {omega2:.3f} ({m9.interpret(omega2, 'eta2')})")
```

You will see η^2 in most published work, and partial η^2 in software output. For a one-way design, partial η^2 and η^2 are the same number.

Report both, interpret one

Give η^2 so readers can compare with the literature, and describe the magnitude from ω^2 .

Significant, and Small



What the effect size is for

The test was significant. The three distributions are nearly the same shape in nearly the same place. Knowing someone's age band tells you almost nothing about their optimism — and only the effect size says so.

Step 6: Generate the Report

```
# Step 6. Every number in the sentence comes from the data and the result,  
# both degrees of freedom included - Welch's second one is not n - k.  
m9.report_anova(survey, "toptim", "agegp3", welch,  
                outcome_label="Total Optimism", order=ages)
```

The sentence carries

- Each group's n , mean and SD, in your order
- F with **both** degrees of freedom, and p
- η^2 and ω^2 , with the magnitude in words
- What the test does not establish

Then the post-hoc sentence

Generate that too — the notebook builds it from the Games–Howell table in a few lines. Nothing in either sentence is typed.

A p -Value of 1.245

An ANOVA report: “ $F(2, 2, 227) = 1.24, p = 1.245$ ”.

What happened

- The sentence was a template with slots for the numbers
- A typed “2” sat in front of two generated degrees of freedom
- The p slot was filled with the variable holding F ; the computed p was never used

Why it survived

- Every number came from a variable, so it *looked* generated
- Nobody asked whether F can have three degrees of freedom
- Nobody noticed a probability larger than one

Generated is necessary, not sufficient

Generate from the result object with a helper that has been tested, and still read the sentence: two degrees of freedom, and $0 \leq p \leq 1$.

Break — 30 minutes: Run the ANOVA

20 minutes on the analysis, then 10 for tea.

In `notebook.ipynb`

- **Part 1**, all six steps, top to bottom
- Read each output before running the next cell

Then in `exercise.ipynb`

- Question 1 — self-esteem across the same three age groups

Think about

Compare the result with the prediction you made at the stand-up. Which part did you get wrong — the direction, the pairs, or the size?

Then stop

The last ten minutes are tea, not catch-up. If Part 1 did not finish, bring it to the end of the session.

Where You Should Be: the ANOVA

You should now have

- A Welch's F with **two** degrees of freedom, the second not a whole number
- A Games–Howell table, read row by row, and a generated sentence for it
- ω^2 described in words, and a conclusion that says what the evidence does *not* establish

The two that catch people

The group order written out — not left to sorting. And the variance test *described*, never used to pick the ANOVA or the post-hoc test.

Many Comparisons

When the Omnibus Finds Nothing

Sleepiness across three age groups, in a staff sleep study: no evidence of any difference.

Then

- **No post-hoc tests.** There is nothing to locate
- Still report the effect size — it says how large a difference the data are consistent with — and what the result does not establish
- Write the age order out: ≤ 37 sorts after the digits, so the youngest band prints last

Do not go hunting

Running the pairs after a non-significant omnibus, until one falls below .05, is the many-*t*-tests problem with an extra step in front.

The Multiple Comparison Problem

Not only an ANOVA problem: any set of tests on one dataset.

- Every test at .05 has a one-in-twenty chance of a false positive
- Twenty tests of true null hypotheses: expect one “finding”
- The more you look, the more you find, whether or not anything is there

The fix

Make each test's threshold stricter, so that the chance of *any* false positive across the family stays at .05.

Watch out

The family is every comparison you made, not every comparison you reported. Leaving the non-significant ones out of the write-up does not shrink it.

Bonferroni, and Holm

Bonferroni

- Divide α by the number of tests m — or, equivalently, multiply each p by m
- Ten tests: each must reach $.05/10 = .005$
- Simple, and strict: as m grows it misses real differences (Type II errors)

Holm

- Sort the p -values; multiply the smallest by m , the next by $m - 1$, and so on, stopping at the first that fails
- Protects against the same error as Bonferroni
- Never less powerful — so there is little reason not to use it

Either is defensible. No correction is not.

Name the one you used: a post-hoc p -value means nothing until the reader knows how it was adjusted.

Planned, Post Hoc, and Unreported

Planned comparisons

- A small number, chosen before the data were seen, from the research question
- Need less correction — because there are fewer of them, and because they were not chosen by looking

Post-hoc comparisons

- Every pair, after the omnibus test
- Corrected for all of them, whichever turn out interesting

Reading someone else's study

You can only judge a multiple-comparison correction if you know how many comparisons were made. A paper that reports only its significant differences, without the total, cannot be evaluated. Plan the analyses before the data, and report all of them.

When Normality Fails

Kruskal–Wallis: Ranks for Three or More

The rank-based counterpart of one-way ANOVA — Mann–Whitney, extended to k groups.

Reach for it when

- The outcome is ordinal
- It is clearly skewed or bounded
- The groups are too small to see their shape

It still assumes

- Independent observations, each counted once
- An outcome that can at least be ranked
- For a statement about *medians*: groups with similar shapes

Pool, rank, compare

Rank every observation in one list, regardless of group. If the groups do not differ, each group's average rank should be close to the overall average rank.

The H Statistic

$$H = \frac{12}{N(N+1)} \sum_{j=1}^k \frac{R_j^2}{n_j} - 3(N+1)$$

- R_j — the sum of the ranks in group j
- n_j — that group's size; N — everyone
- Large H : the groups' ranks are further apart than chance explains
- Compared against χ^2 on $k - 1$ degrees of freedom
- Corrected for ties, which questionnaires and grades are full of
- Reported as $H(k - 1)$; some software labels it “chi-squared”

The same idea as Unit 4

With two groups, Kruskal–Wallis is Mann–Whitney by another route: the same comparison of ranks, differing only in the small-sample details.

A Worked Question: Grades and Social Class

The Youth Cohort Study of England and Wales: thirteen thousand school-leavers, their GCSE maths grade and their family's social class.

Hypotheses

H₀: The distribution of maths grades is the same in every social class group.

H₁: In at least one group, maths grades tend to be higher or lower than in the others.

gradmath: A* to G, then Fail — ordinal.

famsec: six groups, five of them rungs of a class scale and one labelled *Other*.

Watch out

Read your H₁ back. Written as “there is *no* difference”, it is H₀ twice — and then no result can support it.

Stand Up — 5 minutes: Halfway

Five minutes. A new question, and a much bigger dataset.

Food for thought

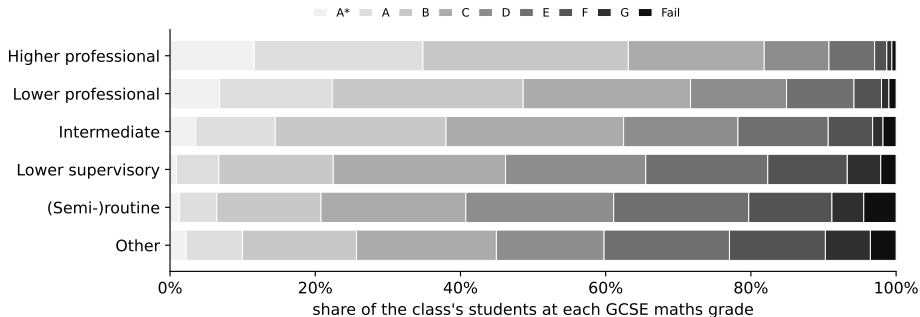
Thirteen thousand students, six groups.

Before seeing anything: will the test be significant? Then the harder one — once you know it will be, what is left for the analysis to tell you?

Still the same six steps

Only step 4 changes: ranks instead of means, Dunn instead of Games–Howell.

Grades Are Ordinal

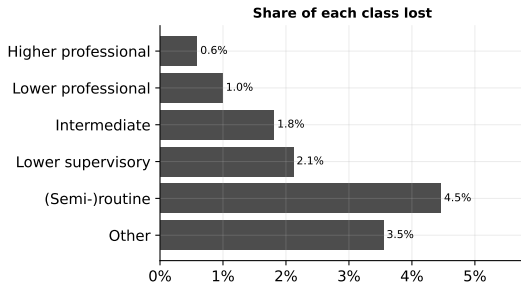
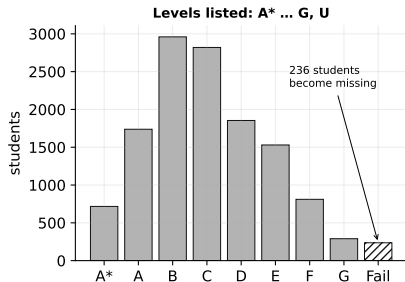


The top grades thin out, and the grades below C spread, at every step down the class scale. *Other* is read as its own group, not as the bottom rung.

Describe the distribution

A mean grade means nothing. Proportions at each grade, medians and quartiles do.

A Recode That Lost the Bottom Grade



The grades were put in order with a list ending in U, as GCSE results print it. This dataset says Fail. Every failed grade became missing — with no error, no warning.

And not at random

The lower the class, the more of it vanished. The analysis understated the very gradient it was looking for.

Recode, Then Count

```
# An ordinal outcome stored as text. Write the order out in full,  
# then check that the recode lost nothing it should not have.  
grades = ["A*", "A", "B", "C", "D", "E", "F", "G", "Fail"]  
youth["mathcode"] = youth.gradmath.map({g: i + 1 for i, g in enumerate(grades)})  
  
lost = youth.gradmath.notna() & youth.mathcode.isna()  
print(f"grades lost to the recode: {lost.sum()}")
```

- Write the order out in full — text sorts A before A*, and Fail between F and G
- Any value the mapping does not name becomes missing

The habit

Count before and after every recode. The numbers must agree, or you must be able to say exactly why they do not.

Running Kruskal–Wallis

```
# Kruskal-Wallis uses only the ranks, so the codes need only be in order.  
# Here 1 is A* and 9 is Fail: a LOW rank is a GOOD grade.  
kw = youth.dropna(subset=["mathcode"])  
by_class = [kw.loc[kw.famsec == c, "mathcode"] for c in classes]  
h = stats.kruskal(*by_class)  
  
eta2_h = m9.eta_squared_h(h.statistic, len(by_class), len(kw))  
print(f"H = {h.statistic:.1f}, p = {h.pvalue:.3g}, eta squared (H) = {eta2_h:.3f}")
```

- Codes need only be in the right order — only their ranks are used
- Here a *low* code is a *good* grade. Say so in the report

Watch out

At this size scipy prints p as zero. It is not zero: report $p < .001$. And with thirteen thousand students, p was never going to be the finding.

Following Up: Dunn's Test

```
# Dunn's test on the same pooled ranks. The correction is not optional:  
# fifteen pairs at .05 would turn up a false positive more often than not.  
pairs = m9.dunn(kw, "mathcode", "famsec", p_adjust="holm", order=classes)  
pairs[["A", "B", "z", "p_unadj", "p_adj"]]
```

Dunn's test

- Compares each pair's mean rank, on the ranks the omnibus test already used
- Six groups: fifteen pairs

The correction is an argument

p_adjust has no default. Holm or Bonferroni are both defensible; which one you used is part of the result.

Effect Size for Kruskal–Wallis

$$\eta_H^2 = \frac{H - k + 1}{N - k}$$

- The rank-based share of variance
- Read against the same benchmarks as η^2 : .01/.06/.14
- This is what R's `rstatix::kruskal_effsize` reports

At thirteen thousand

Every p is tiny. The effect size, and the pattern in the grades, are the finding: how strongly class and grades travel together, and in which direction.

What Kruskal–Wallis Claims

It compares whole distributions of ranks, not medians.

A significant result supports

“Grades in some class groups tend to be higher than in others.”

It does not, on its own, support

“The median grade differs between class groups.”

(That reading needs the distributions to have a similar shape.)

Report medians and quartiles *as description*, and phrase the conclusion as a tendency. And it is an association: social class travels with school, region and income, and the test cannot separate them.

Break — 30 minutes: Nothing, Then Ranks

25 minutes on the analyses, then 5 to stretch.

In `notebook.ipynb`

- **Part 2** — the sleep study, where nothing is found
- **Part 3** — grades and social class, recode first

Then in `exercise.ipynb`

- Questions 2 and 3, as far as you get

Part 2 is short

It is there to show what you do *not* do after a non-significant omnibus test. Do not skip it to reach the ranks.

Think about

Question 2 has a group of two people in it. What can a group of two tell you — and what does it do to a Welch test?

Where You Should Be: Ranks and Corrections

You should now have

- A non-significant ANOVA reported with its effect size, and no post-hoc tests after it
- A recode checked by counting, before any test was run on it
- A Kruskal–Wallis result with η_H^2 , and a Dunn table whose correction you can name

If something did not run

Keep going — the reporting section needs none of it, and the worked solution is released after this class. Note where it stopped and ask.

Reporting

How to Report a Multi-Group Comparison

Everything from Unit 4, plus the follow-up.

- What test, on what variables, with how the groups were formed
- Descriptives per group, with n — means and SDs, or medians and quartiles for ranks
- The statistic, **both** degrees of freedom for F , and p
- An effect size, **interpreted**
- The post-hoc test, its correction, and which pairs differed
- A conclusion referring back to the hypothesis

Convention

p to three decimals, no leading zero; below .001, write $p < .001$.

The effect size **even when nothing is significant**, and the post-hoc comparisons **only when the omnibus test was**.

The Rest of the Family

Two-way ANOVA

- Two factors at once — age band *and* sex
- Adds an **interaction**: does one factor's effect depend on the other?

Repeated measures

- The same people under every condition
- Repeated-measures ANOVA; Friedman's test for ranks
- *Mixed* designs combine both

Beyond

- ANOVA is a linear model with categorical predictors
- Regression, from Unit 6, is the same machinery

The design decides, again

Which member of the family you need is settled by how the data were collected — between people, within people, or both — before any of it is analysed.

Before Step 4: Preparing the Data

Every preparation decision from Unit 3 is made before the test, and changes what the test sees.

- **Weights** — the cohort data carry one that scipy's rank test cannot apply: say the analysis is unweighted
- **Missing data** — on the model variables, and why
- **Outliers** — found, explained, not silently dropped
- **Recoding** — written out, then counted
- **Transforming** — only if it helps, and then you interpret the transformed scale
- **Or rank** — if a transformation does not fix the shape, use a test that does not need it

Watch out

Every one of these belongs in the write-up. A reader cannot check a decision they were not told about.

Two Conclusions You Must Not Draw

A significant omnibus test

Does *not* mean every group differs from every other. It means the means are not all equal; the post-hoc comparisons say which differ — and often it is fewer than you expect.

A non-significant pair

Does *not* mean those two groups are the same. It means this sample, under this correction, did not provide sufficient evidence of a difference between them.

What you should take from this week

You SHOULD be able to

- Say why many t -tests inflate the familywise error rate
- Explain how F weighs variation between groups against within
- Check assumptions within groups; describe spread, never gate on it
- Run Welch's ANOVA with Games–Howell, and Kruskal–Wallis with Dunn under a named correction
- Report ω^2 , η^2 or η_H^2 , with the magnitude in words
- Recode an ordinal variable and prove by counting that nothing was lost

This week: the notebook works both tests end to end; the exercise asks you to do the same on questions you have not seen.

That's all for this week